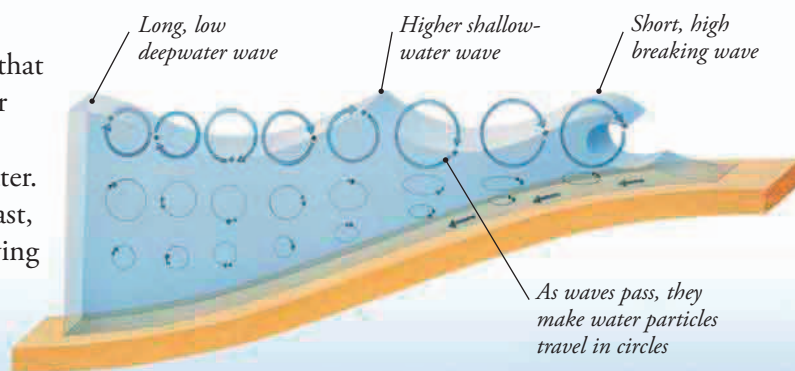


Waves and currents

Oceanic winds whip up waves on the ocean surface, and drive currents that swirl around the oceans in vast whirlpools called gyres. These surface currents are linked to deepwater currents. Together, they form a system that carries ocean water all around the world.

RIPPLES AND WAVES

As the wind blows over the ocean, it creates ripples that grow into waves. The farther waves travel, the bigger they get, so strong winds can generate huge waves that get steeper as they push into shallow coastal water. But even though waves travel over the water quite fast, the water itself stays in roughly the same place, moving very slightly forward as each wave passes.



SHIP SINKERS

In some places, opposing waves may meet head-on. They can either cancel each other out or combine to create huge "rogue waves" up to 100 ft (30 m) high—big enough to break right over ships and even sink them. Several big ships that have disappeared at sea were probably hit by such waves.

